

FinRisk: Intelligent Credit & Fraud Engine

Driving Faster, Safer Lending and Real-Time Fraud Prevention for **Metropolitan Banking Group**

Key Highlights:

Client: Metropolitan Banking Group (MBG), UK Financial Services

Focus: Credit Risk & Fraud Management in Banking & Fintech

Scope: 50k+ credit applications/month; 2M+ transactions/month

Challenges Addressed: Reducing 3.2% default rate & £2.1M annual fraud losses



The Current Landscape: Pressing Challenges

1

Suboptimal Credit Decisions

18% False Positives (rejected good customers)

12% False Negatives (approved bad customers)

2

Manual & Slow Processes

5–7 days manual underwriting per application

- High review burden for 'thin-file' customers

3

Fraud Operations Overload

2,000+ false alerts daily

- Reactive rather than proactive prevention

4

Limited Risk Visibility

- Absence of portfolio stress testing capabilities
- Insufficient real-time risk monitoring

5

Regulatory Demands

- Mandatory explainable models for all decisions
- Comprehensive audit trails required

Our Mission for the Next 4 Weeks

FinRisk: Objectives & Scope

Key Technical Deliverables:

Credit Scoring: Develop explainable XGBoost and Logistic Regression models (SHAP/LIME)

Fraud Detection: Implement anomaly models (Isolation Forest, One-Class SVM)

Feature Engineering: Extract powerful behavioral & credit bureau signals

Strategic Outcomes & Deployment:

Portfolio Analytics: Enable stress testing and enhanced risk monitoring

Production API: Deploy a FastAPI service for <100ms scoring, with built-in monitoring

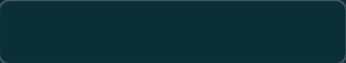
Governance: Ensure full model lifecycle management



Core Data Assets & Model Store

Leveraging MBG's rich historical data for robust model development and transparent tracking.

Application Data	100k+ loan applications (24 months historical, labeled outcomes)
Transaction Data	5M+ customer transactions (0.1% confirmed fraud rate)
Customer Profiles	Demographics & behavioral history (3+ years)
Credit Bureau Data	Comprehensive external credit assessment data
Model Store Tables	<code>credit_applications, transaction_data, customer_profiles, credit_bureau_data, model_predictions</code>
Data Model Logic	- One customer linked to many applications/transactions - Predictions table captures scores, features, explanations, and full audit trails



Robust Architecture & Tech Stack



Modeling Libraries

scikit-learn, XGBoost, LightGBM, statsmodels



Data Processing

pandas/NumPy, Dask for scalable computation



Model Serving

FastAPI, PostgreSQL, Redis for high-speed caching



Monitoring & MLOps

MLflow, Weights & Biases, custom KPI dashboards



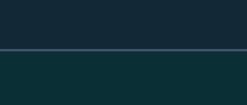
Deployment

Docker, AWS EC2/Heroku for flexible deployment



End-to-End Flow

Ingest → Feature Engineering → Train/Validate → Register (MLflow) → Serve (API+Redis) → Monitor (latency, drift, accuracy)



Model Strategy & Validation Rigour

Our approach combines cutting-edge machine learning with rigorous validation to ensure effective, transparent, and compliant models for credit risk and fraud detection.



Credit Risk Models

Primary: XGBoost for Predictive Power. Superior accuracy (+5% AUC-ROC uplift) and ability to handle complex non-linear relationships.

Optimized via GridSearchCV.

Baseline: Logistic Regression for Interpretability & Benchmark. Provides a robust, transparent baseline essential for regulatory compliance and performance comparison.

Feature Importance: SHAP values for systematic analysis of feature contributions, providing actionable insights into creditworthiness drivers (e.g., income, debt-to-income, payment history).



Fraud Detection Models

Anomaly Detection: Isolation Forest & One-Class SVM. Unsupervised techniques to identify suspicious patterns in high-dimensional transaction data without pre-labeled cases.

Threshold Calibration: Optimized for Precision/Recall Balance. Aims for 95% recall on confirmed fraud with <0.5% false positive rate to minimize operational burden.

Real-time Adaptation: Sub-100ms prediction times for immediate alerts. Explores continuous retraining (e.g., weekly) to adapt to emerging fraud patterns.

Key Success Metrics & Targets

Clear, measurable targets ensuring the FinRisk engine delivers tangible business value and regulatory compliance.

>0.75

Credit AUC

Area Under Curve

>0.60

Fraud Precision

True Positives / (True Positives +
False Positives)

>0.80

Fraud Recall

True Positives / (True Positives +
False Negatives)

<100ms

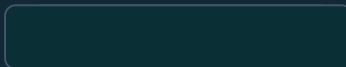
API Latency

For real-time scoring

<15%

Credit False Positives

Reducing erroneous rejections



Monitoring, Governance & Risk Controls

Establishing a resilient framework for continuous oversight, compliance, and proactive risk mitigation.

Real-Time Monitoring

- KPIs: Accuracy, data drift (PSI/KL divergence)
- Alert volumes & operational review load
- Automated data quality checks on inputs

Governance & Audit

- Versioning: Models, features, datasets tracked
- Comprehensive decision logs for every prediction
- Bias & fairness dashboards with remediation workflows

Risk Mitigations

Class Imbalance: SMOTE, cost-sensitive learning

Black-Box Models: SHAP/LIME for explainability

Model Drift: Continuous monitoring & automated retraining

Latency: Caching, data compression, parallel processing

Key Deliverables & 4-Week Timeline

A clear roadmap to achieve our objectives within the agreed timeframe, culminating in a fully operational FinRisk engine.

